Enrolment No. _____

MID TERM EXAMINATION

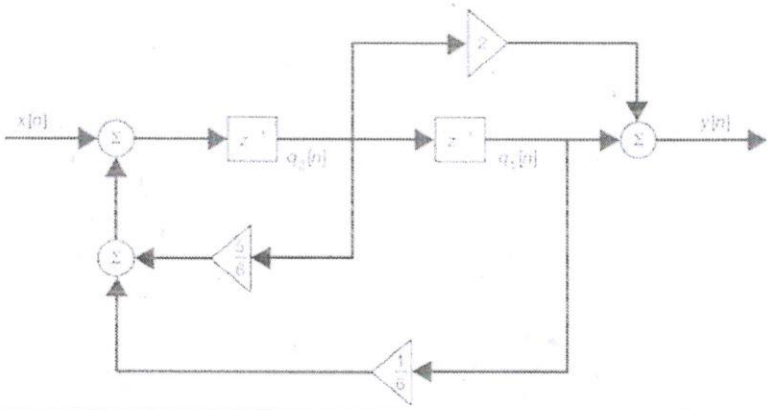
B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

CSE/IT- 4th Semester, April, 2024

Paper Code: EEC-208 **Subject: Circuits and Systems**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Check linearity and time-invariance of system y(n) = 2n+x(n).	2	CO1
1 (b)	Explain an overdamped response of series RLC circuit and sketch the curve.	2	CO2
1 (c)	Find the Laplace transform and associated ROC of signal $x(t) = e^{-2t}$ [u(t) - u(t - 5)].	3	CO1
1 (d)	For the LTI system described by the following equation. Determine the impulse function of system. $y''(t) + y'(t) - 2y(t) = x(t)$	3	CO2
2 (a)	Given $X(z) = \frac{z(z-4)}{(z-1)(z-2)(z-3)}$ (i) State all the possible regions of convergence. (ii) For which ROC, z-transform X(z) is causal sequence	6	CO1
2 (b)	Find the state space representation of the discrete-time system with the state variables $q_1[n]$ and $q_2[n]$ shown in Figure. 	4	CO1
3 (a)	Obtain expression for current i(t) and voltage across the inductor in the circuit shown in figure.	4	CO2

PTO